

Module -2

Step Response of Series RLC Circuits

Dr. Sathya P
Associate Professor
SENSE

Step response of a Series RLC Circuit

The step response is obtained by the sudden application of a dc source.

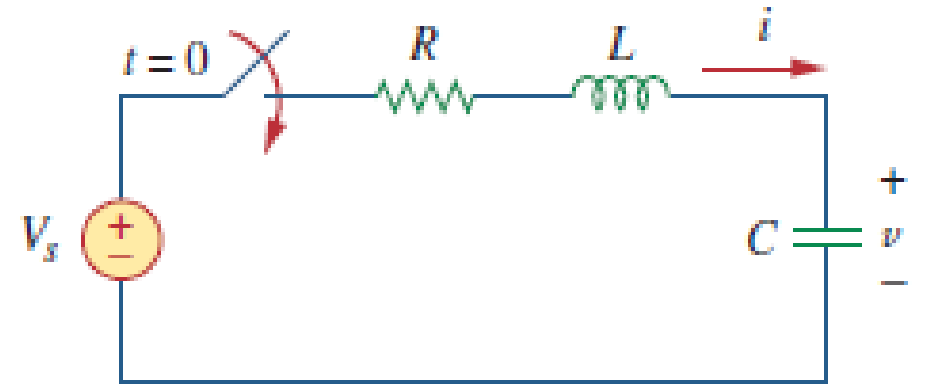
Applying KVL around the loop for $t > 0$,

$$L \frac{di}{dt} + Ri + v = V_s \longrightarrow (1)$$

$$i = C \frac{dv}{dt} \longrightarrow (2)$$

Substituting for i

$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC} \longrightarrow (3)$$



$$v(t) = v_t(t) + v_{ss}(t)$$

$$v_{ss}(t) = v(\infty) = V_s$$

$$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v_t(t) = (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

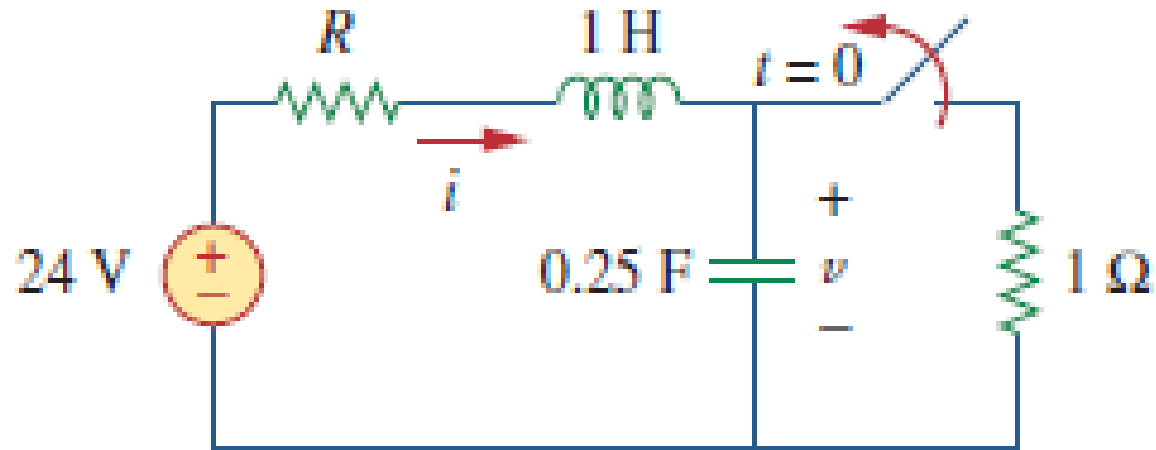
$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

Numerical - 1

Find $v(t)$ and $i(t)$ for $t > 0$. Consider these cases: $R = 5\ \Omega$, $R = 4\ \Omega$, and $R = 1\ \Omega$,



Solution:

■ **CASE 1** When $R = 5\ \Omega$. For $t < 0$, the switch is closed for a long time. The capacitor behaves like an open circuit while the inductor acts like a short circuit. The initial current through the inductor is

$$i(0) = \frac{24}{5 + 1} = 4\text{ A}$$

and the initial voltage across the capacitor is the same as the voltage across the $1\text{-}\Omega$ resistor; that is,

$$v(0) = 1i(0) = 4\text{ V}$$

For $t > 0$, the switch is opened, so that we have the $1\text{-}\Omega$ resistor disconnected. What remains is the series RLC circuit with the voltage source. The characteristic roots are determined as follows:

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2$$
$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1, -4$$

Since $\alpha > \omega_0$, we have the overdamped natural response. The total response is therefore

$$v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$$

$$v(t) = 24 + (A_1 e^{-t} + A_2 e^{-4t})$$

We now need to find A_1 and A_2 using the initial conditions.

$$v(0) = 4 = 24 + A_1 + A_2$$

$$-20 = A_1 + A_2$$

The current through the inductor cannot change abruptly and is the same current through the capacitor at $t = 0^+$ because the inductor and capacitor are now in series. Hence,

$$i(0) = C \frac{dv(0)}{dt} = 4 \quad \Rightarrow \quad \frac{dv(0)}{dt} = \frac{4}{C} = \frac{4}{0.25} = 16$$

$$\frac{dv}{dt} = -A_1 e^{-t} - 4A_2 e^{-4t}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 16 = -A_1 - 4A_2$$

$$A_1 = -64/3 \text{ and } A_2 = 4/3.$$

$$v(t) = 24 + \frac{4}{3}(-16e^{-t} + e^{-4t}) \text{ V}$$

Since the inductor and capacitor are in series for $t > 0$, the inductor current is the same as the capacitor current. Hence,

$$i(t) = C \frac{dv}{dt}$$

$$i(t) = \frac{4}{3}(4e^{-t} - e^{-4t}) \text{ A}$$

Note that $i(0) = 4 \text{ A}$, as expected.

■ **CASE 2** When $R = 4\ \Omega$. Again, the initial current through the inductor is

$$i(0) = \frac{24}{4 + 1} = 4.8\ \text{A}$$

and the initial capacitor voltage is

$$v(0) = 1i(0) = 4.8\ \text{V}$$

For the characteristic roots,

$$\alpha = \frac{R}{2L} = \frac{4}{2 \times 1} = 2$$

while $\omega_0 = 2$ remains the same. In this case, $s_1 = s_2 = -\alpha = -2$, and we have the critically damped natural response. The total response is therefore

$$v(t) = v_{ss} + (A_1 + A_2 t)e^{-2t}$$

and, as before $v_{ss} = 24\ \text{V}$,

$$v(t) = 24 + (A_1 + A_2 t)e^{-2t}$$

To find A_1 and A_2 , we use the initial conditions. We write

$$v(0) = 4.8 = 24 + A_1 \quad \Rightarrow \quad A_1 = -19.2$$

Since $i(0) = C dv(0)/dt = 4.8$ or

$$\frac{dv(0)}{dt} = \frac{4.8}{C} = 19.2$$

$$\frac{dv}{dt} = (-2A_1 - 2tA_2 + A_2)e^{-2t}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 19.2 = -2A_1 + A_2$$

$$A_1 = -19.2 \text{ and } A_2 = -19.2.$$

$$v(t) = 24 - 19.2(1 + t)e^{-2t} \text{ V}$$

The inductor current is the same as the capacitor current; that is,

$$i(t) = C \frac{dv}{dt}$$

$$i(t) = (4.8 + 9.6t)e^{-2t} \text{ A}$$

$$\therefore i(0) = 4.8 \text{ A,}$$

■ **CASE 3** When $R = 1 \Omega$. The initial inductor current is

$$i(0) = \frac{24}{1 + 1} = 12 \text{ A}$$

and the initial voltage across the capacitor is the same as the voltage across the $1\text{-}\Omega$ resistor,

$$v(0) = 1i(0) = 12 \text{ V}$$

$$\alpha = \frac{R}{2L} = \frac{1}{2 \times 1} = 0.5$$

Since $\alpha = 0.5 < \omega_0 = 2$, we have the underdamped response

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -0.5 \pm j1.936$$

The total response is therefore

$$v(t) = 24 + (A_1 \cos 1.936t + A_2 \sin 1.936t)e^{-0.5t}$$

We now determine A_1 and A_2 . We write

$$v(0) = 12 = 24 + A_1 \quad \Rightarrow \quad A_1 = -12$$

Since $i(0) = C dv(0)/dt = 12$,

$$\frac{dv(0)}{dt} = \frac{12}{C} = 48$$

But

$$\begin{aligned} \frac{dv}{dt} &= e^{-0.5t}(-1.936A_1 \sin 1.936t + 1.936A_2 \cos 1.936t) \\ &\quad - 0.5e^{-0.5t}(A_1 \cos 1.936t + A_2 \sin 1.936t) \end{aligned}$$

At $t = 0$,

$$\frac{dv(0)}{dt} = 48 = (-0 + 1.936A_2) - 0.5(A_1 + 0)$$

Substituting $A_1 = -12$ gives $A_2 = 21.694$,

$$v(t) = 24 + (21.694 \sin 1.936t - 12 \cos 1.936t)e^{-0.5t} \text{ V}$$

$$i(t) = C \frac{dv}{dt}$$

$$i(t) = (3.1 \sin 1.936t + 12 \cos 1.936t)e^{-0.5t} \text{ A}$$

